

Zhenting Zhu

☎ (+1)424-430-5498 | ✉ u007zzt@gmail.com/whzzt@ucla.edu | 🌐 whzzt

EDUCATION

- **Ph.D. Program, University of California, Los Angeles** Los Angeles, United States of America
UCLA Samueli School of Engineering
Sep. 2023 – Present
◦ Second-year Ph.D student advised by Prof. [Harry Xu](#).
- **Undergraduate Program, Tsinghua University** Beijing, China
Yao Class, Institute for Interdisciplinary Information Sciences (IIIS)
Aug. 2019 – Jun. 2023
◦ Enrolled in Yao class, a special pilot CS program led by Prof. [Andrew C. Yao](#) for talented students.
◦ Granted to Tsinghua as the 3rd place in International Olympiad in Informatics 2018 as a member of Chinese National Team.
◦ Overall GPA: 3.83/4.00, Major GPA: 3.96/4.00.

RESEARCH INTERESTS

My primary research interest lies in building resilient and intelligent systems for modern applications. I focus on key challenges within large-scale distributed platforms, spanning both conventional data center architectures and the specialized infrastructure required for high-performance machine learning. My work specifically investigates how to enhance system resilience, usability, and reliability in these demanding environments.

PUBLICATIONS

- **Orthrus: Efficient and Timely Validation of Silent User Data Corruption on the Cloud** SOSP' 25
Chenxiao Liu*, **Zhenting Zhu***, Quanxi Li, Yanwen Xia, Yifan Qiao, Xiangyun Deng, Youyou Lu, Tao Xie, Huimin Cui, Zidong Du, Harry Xu, Chenxi Wang
◦ *Silent data corruption (SDC)* is one particular category of CPU errors transpires discreetly, without crashing applications or triggering hardware warnings. These elusive errors pose a significant threat by undermining user data. In this paper, we proposed Orthrus, a solution that provides timely detection of SDC caused by post-installation CPU errors with low overhead and minimal engineering effort.
- **VQPy: An Object-oriented Video Query Language** (Project Repo) MLSys' 24
Shan Yu, **Zhenting Zhu**, Yu Chen, Hanchen Xu, Pengzhan Zhao, Yang Wang, Arthi Padmanabhan, Hugo Latapie, Harry Xu
◦ Video analytics is widely used in contemporary systems and services. At the forefront of video analytics are video queries that users develop to find objects of particular interest. In this paper, We presented an object-oriented, open-source framework named VQPy specialized in video analytics. Our main contributions include the design and implementation of a video-object-oriented frontend, an efficient backend with an object-centric data model, and an extensible optimization framework.

RESEARCH EXPERIENCES

- **Ph.D. Student, UCLA SOLAR Lab** (Advisor: [Harry Xu](#)) University of California, Los Angeles
Resumed since offer acceptance in Mar. 2023 Sep. 2023 – Present
- **Research Intern, UCLA SOLAR Lab** (Advisor: [Harry Xu](#)) University of California, Los Angeles
Full-time, Onsite Jul. 2022 – Sep. 2022 Feb. 2022 – Feb. 2023
- **Research Intern, Tsinghua IDEAL Lab** (Advisor: [Mingyu Gao](#)) Tsinghua University
Part-time Sep. 2021 – Jan. 2022 Sep. 2021 – Jan. 2022

SELECTED HONORS AND AWARDS

- **2025 ICPC North America Championship, Bronze Medal** ICPC Committee
As a member of the team UCLA Three Musketeers, ranked 4th among 52 teams that advanced from regionals May. 2025
- **Yao Class summa cum laude** Tsinghua University
Awarded to outstanding fresh graduates in IIIS (Top 25%) Jun. 2023
- **Yao Award Bronze Medal** (Introduction) Tsinghua University
Recognize the outstanding students in Yao class (Top 10%) Sep. 2022
- **The 44th annual ICPC World Final, Invitational Silver Medal & High Honors** ICPC Committee
Remote because of the epidemic, ranked 3rd among 56 teams that advanced from all over the world Oct. 2021
- **ICPC Asia East Continent Final 2019/2020, Champion** ICPC Committee
As a member of the Tsinghua University team Three Hold Two Dec. 2019 / Apr. 2021
- **CCPC Final 2019/2020, Champion** CCPC Committee
As a member of the Tsinghua University team Three Hold Two Nov. 2019 / May. 2021
- **First Class Freshman Scholarship** Tsinghua University
Awarded to about 1% of students Oct. 2019
- **International Olympiad in Informatics 2018, Gold Medal** IOI Organization
Ranked 3rd among 335 participants from 87 countries Sep. 2018
- **Chinese National Olympiad in Informatics 2017, Gold Medal** China Computer Federation
Ranked 29th among 283 participants Jul. 2017